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Autonomous robotic ultrasound scanning on complex surfaces: Automatic trajectory generation and disturbance observer-based model predictive admittance control

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ABSTRACT

Robotic ultrasound scanning facilitates clinical diagnosis by reducing operator burden and enhancing imaging reproducibility. To achieve stable human-robot interaction on complex anatomical surfaces, this paper proposes an autonomous compliant ultrasound scanning framework. First, an equidistant trajectory planning method based on 3D point clouds is presented to satisfy probe pose constraints. By establishing an anisotropic geodesic distance field to characterize the surface topology of the human body, this approach achieves uniform global coverage and avoids abrupt orientation shifts. Second, a disturbance observer-based model predictive admittance control (DO-MPAC) scheme is developed to maintain a stable and compliant interaction force. Within this scheme, environmental uncertainties are estimated and compensated for through an extended state observer (ESO), and the dependency of the force error dynamics on the admittance parameters is mitigated through the design of a virtual control law. Furthermore, a predictive optimization framework is utilized to guarantee optimal force tracking. Finally, simulations and physical experiments evaluate the proposed system against conventional, adaptive, and sliding mode admittance controllers. Results demonstrate that in complex curved environments, DO-MPAC achieves a faster transient response and higher steady-state precision, reducing the maximum force deviation (MFD), integral of absolute error (IAE), and integral of time-weighted absolute error (ITAE) by at least 88.1%, 45.6%, and 43.0%, respectively.

1. Introduction

Benefiting from its radiation-free, cost-effective, and real-time imaging capabilities, ultrasound has become an essential and widely used imaging technique in clinical diagnosis (Moore & Copel, 2011; Priester et al., 2013). However, conventional manual ultrasound examinations rely heavily on the sonographer's expertise, which often leads to fatigue during long-term operations. Additionally, the subjective nature of manual scanning frequently results in high inter-operator variability and poor image reproducibility (Evans et al., 2009; Harrison & Harris, 2015). To overcome these practical limitations, extensive research efforts have been devoted to the development of robot-assisted ultrasound systems (Jiang et al., 2023; Ning et al., 2021). It is worth noting that during continuous ultrasound scanning, the probe-tissue interface is not a simple rigid contact, but rather a complex interactive process characterized by soft tissue deformation, surface stiffness variations, and external disturbances (Sigrist et al., 2017). For robotic ultrasound to obtain stable, clear, and reproducible images, it is crucial to maintain the probe perpendicular to the tissue surface while simultaneously regulating a safe, compliant, and constant contact force (Ihnatsenka & Boezaart, 2010).

To guarantee a perpendicular posture and a steady contact force between the probe and the tissue, numerous studies have focused on scanning trajectory planning and constant force control (Fu et al., 2022; Su et al., 2024). In practice, the human body surface presents a complex and dynamic physical environment. Its spatial and stiffness features vary continuously depending on anatomical regions, individual differences, and physiological activities, which complicates the execution of precise and robust force control during interaction (Cao et al., 2025). In addition, the quality of compliant force control is directly determined by the pre-planned generated scanning trajectory, namely, the pose sequence of the ultrasound probe. If the planned trajectory is suboptimal, particularly in areas with high curvature or abrupt transitions, the probe will experience sharp pose shifts and discontinuous motions. During

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physical contact, these movement errors directly translate into force fluctuations, local instability, or even imaging artifacts (Von Haxthausen et al., 2021). Therefore, achieving an ideal probe-tissue interaction requires not only efficient global trajectory generation but also robust compliant force control.

Trajectory planning aims to determine the optimal probe pose, and automatic trajectory generation has become a vital development trend in the field of medical robotics. Early ultrasound robots involved frequent manual intervention, where trajectory points were specified on a 2D reference plane followed by linear interpolation (Şen et al., 2016; Su et al., 2021). However, these predefined 2D paths fail to adapt to the 3D curvature variations of the human body, often resulting in normal pose deviations of the probe and loss of acoustic coupling. To address this, several approaches (Welleweerd et al., 2020; Zhan et al., 2020) mapped predefined 2D base routes onto 3D surfaces. Although this strategy improves surface adherence to some extent, it struggles to ensure complete 3D scanning coverage. Advancing further, the methods proposed in Hennersperger et al. (2016), Jiang et al. (2022), Virga et al. (2016) planned 3D paths using preoperative medical images and registered them to the patient's body surface point cloud. Nevertheless, these semi-automatic techniques rely on offline data and human involvement. To eliminate the need for prior data and realize autonomous scanning, recent research has focused on direct trajectory planning based on 3D point clouds (Tan et al., 2023, 2022; Wang et al., 2022; Zhang et al., 2025). Specifically, an equidistant 3D point cloud search algorithm was designed in Wang et al. (2022) to produce a fully covered circular scanning trajectory, but its massive computational cost and unidirectional progression strategy often result in local discontinuities. Through parallel slicing of large-scale point clouds and curve fitting, the method in Tan et al. (2022) automatically generated smooth skeleton paths that cover the global region. Based on this, the work in Zhang et al. (2025) proposed a bi-objective optimization algorithm that balances acceleration continuity and position deviation to correct local geometric anomalies and improve pose continuity. Despite these advancements, these slicing methods face an inherent trade-off between topological feature extraction accuracy and computational burden. Sparse slicing tends to neglect local details, while dense slicing incurs significant computational overhead due to intersection calculations among massive point clouds. Therefore, it remains a critical bottleneck to automatically and efficiently generate a scanning trajectory that adapts to complex surfaces with smooth poses and uniform coverage.

The primary objective of force control strategies is to improve safe and stable constant force regulation at the probe-tissue interface. For most position-controlled medical robotic systems, admittance control (AC) achieves the desired force tracking by adjusting position commands (He et al., 2020; Jiang et al., 2020; Keemink et al., 2018). Classical AC laws have been widely applied across various robotic platforms (Li et al., 2018; Tan et al., 2022). For instance, the method in Tan et al. (2022) utilized a standard admittance controller in an ultrasound robotic system to maintain basic compliant surface contact. However, traditional fixed-parameter admittance schemes struggle to adapt to the dynamic position and stiffness of complex human surfaces, which inevitably causes significant force fluctuations. To enhance force tracking performance in dynamic environments, various improved schemes have been proposed (Duan et al., 2018; Hamedani et al., 2021; Jiang et al., 2024; Li et al., 2026a,b; Yoon et al., 2024). Specifically, adaptive schemes perform parameter tuning based on tracking errors to attain compliant tracking (Duan et al., 2018; Jiang et al., 2024). Nevertheless, these adaptation mechanisms are driven by error accumulation and suffer from phase lag. To overcome this delay, active observation and feedforward-based AC methods (Li et al., 2026a,b) yield rapid responses through online estimation and active compensation of unknown disturbances. Even so, these approaches are limited to instantaneous compensation of current states and lack the ability to predict future dynamic trends. Additionally, the traditional linear sliding surface employed in Li et al. (2026a) introduces the inherent chattering defect of sliding mode control (SMC). To handle nonlinear interaction dynamics, controllers based on neural networks have been developed to approximate these dynamics online and enable adaptive tuning of admittance parameters (Hamedani et al., 2021; Yoon et al., 2024). Unfortunately, these online learning models impose an immense computational burden. Notably, all the aforementioned force control schemes often induce a certain degree of overshoot when pursuing rapid response, which poses critical safety hazards during robotic ultrasound scanning. Therefore, an unresolved challenge is to rapidly track the desired force in dynamic environments while suppressing unsafe transient overshoots and maintaining steady-state force stability.

Based on the preceding analysis, an autonomous and compliant robotic ultrasound scanning system is developed in this paper. The main contributions of this work are summarized as follows:

1) An autonomous compliant ultrasound scanning framework tailored for complex environments is proposed. Compared with existing robotic ultrasound systems (Li et al., 2026a; Tan et al., 2022; Welleweerd et al., 2020), the proposed framework integrates an automatic trajectory generation algorithm with a compliant force control strategy within a unified closed-loop architecture. Consequently, it facilitates compliant probe-tissue interaction in dynamic environments and improves the standardization and reproducibility of clinical scans.
2) To satisfy probe pose constraints during scanning, this paper integrates multi-scale mapping and an anisotropic geodesic distance field into robotic ultrasound trajectory planning. Unlike previous approaches (Jiang et al., 2022; Su et al., 2021; Welleweerd et al., 2020), this mechanism eliminates the need for preoperative medical images or manually designed paths. In comparison with conventional methods (Tan et al., 2022; Wang et al., 2022; Zhang et al., 2025), the proposed method suppresses abrupt probe orientation shifts, ensures uniform global coverage, and captures the topological features of complex surfaces with lower computational costs.
3) A disturbance observer-based model predictive admittance control (DO-MPAC) strategy is proposed to ensure stable and compliant contact. Compared with existing probe-tissue force controllers (Jiang et al., 2024; Li et al., 2026a; Tan et al., 2022), this strategy leverages model predictive control (MPC) to proactively mitigate unsafe transient fluctuations. Furthermore, a virtual control law is designed to connect the extended state observer (ESO) and MPC. This formulation suppresses the coupling effect between the tracking error dynamics and the admittance parameters, which alleviates the inherent trade-off between physical compliance and tracking agility.

The remainder of this paper is organized as follows. Section 2 presents the 3D point cloud-driven ultrasound trajectory generation algorithm, and the design process of the proposed DO-MPAC scheme is detailed in Section 3. Numerical simulations under diverse environmental conditions and physical experiments are presented in Sections 4 and 5, respectively. Finally, Section 6 summarizes the conclusions and outlines potential avenues for future research.

2. Automatic generation of anisotropic equidistant scanning trajectories based on 3D point clouds

Fig. 1 illustrates the generation process for ultrasound scanning trajectories derived from 3D point clouds. The proposed autonomous trajectory generation framework facilitates standardized scanning procedures and improves the reproducibility of ultrasound imaging.

Taking lumbar ultrasound scanning as a representative clinical application, this section details the step-by-step trajectory generation procedure.

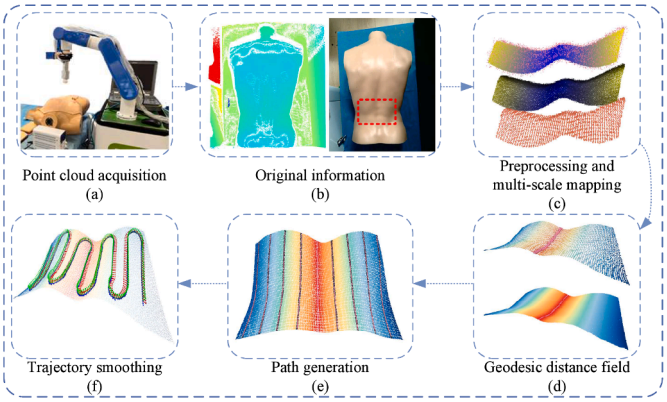


Fig. 1. Automatic trajectory generation workflow based on 3D point cloud.

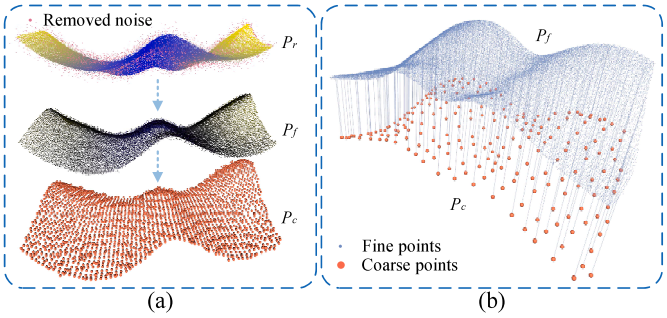


Fig. 2. Preprocessing workflow of point clouds. (a) Raw, fine, and coarse point clouds; (b) Multi-scale point cloud mapping.

2.1. Point cloud acquisition and preprocessing

In clinical scanning environments, the raw 3D point cloud P_r acquired by the RGB-D camera typically suffers from noise and high data density, which stems from hardware precision limitations and human respiratory disturbances. To reduce the time complexity of the algorithm while ensuring trajectory accuracy, this subsection employs a multi-scale spatial discretization strategy.

As shown in Fig. 2(a), voxel filtering and statistical outlier removal are applied to the raw point cloud to obtain the denoised fine-level point cloud P_f (Rusu & Cousins, 2011). Then a secondary downsampling is performed on P_f using a larger voxel size v_{coarse}, thereby extracting the coarse-level landmark point cloud P_c for global distance field computation:

$$P_c = \{\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_N\} \subset \mathbb{R}^3 \tag{1}$$

where N denotes the total number of downsampled points.

Fig. 2(b) illustrates the multi-scale mapping relationship between the fine and coarse point clouds. This multi-scale architecture retains the detailed surface topology in P_f while confining the heavy computational burden of the shortest path search to P_c. Therefore, this approach significantly enhances the computational efficiency of the algorithm.

2.2. Construction of the anisotropic geodesic weight graph

To facilitate uniform trajectory distribution across flat regions and prevent abrupt orientation changes of the ultrasound probe in high-curvature areas, an undirected weighted graph $G = (V, E)$ is constructed on the coarse point cloud P_c.

Specifically, the vertex set V is identical to the coarse point cloud (i.e., $V = P_c$), and the edge set E encodes the local topological connectivity. By integrating local principal component analysis (PCA) with a k-d tree search, the algorithm first estimates the unit normal vector $\mathbf{n}_i$ for each point $\mathbf{p}_i$ in V. Subsequently, the k nearest neighbors in Euclidean space are identified for each vertex to establish the connecting edges $e_{i,j} \in E$. The calculation of local point normals is illustrated in Fig. 3(a).

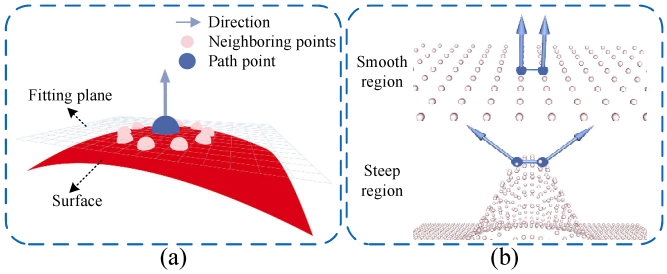


Fig. 3. Feature extraction and surface characterization. (a) Computation of surface normals; (b) Demonstration of anisotropic weights for smooth and steep regions.

To quantify local spatial curvature, the anisotropic geodesic distance weight $w_{i,j}$ between neighboring points $\mathbf{p}_i$ and $\mathbf{p}_j$ is defined as follows:

$$w_{i,j} = \|\mathbf{p}_i - \mathbf{p}_j\|_2 \cdot \left[1 + \alpha\left(1 - |\mathbf{n}_i \cdot \mathbf{n}_j|\right)\right] \tag{2}$$

where $\|\mathbf{p}_i - \mathbf{p}_j\|_2$ represents the base Euclidean distance between the two points. The term $\mathbf{n}_i \cdot \mathbf{n}_j$ is the dot product of their respective normal vectors, whose absolute value characterizes local surface flatness. α denotes the curvature penalty factor, which heavily penalizes abrupt geometric variations by increasing the traversal cost. Fig. 3(b) demonstrates the anisotropic weights for regions with different curvatures. Evidently, the cost in smooth regions approximates the Euclidean distance, whereas it is penalized and amplified in high-curvature areas due to larger normal deviations.

2.3. Geodesic distance field generation

To generate scanning trajectories that completely cover the human surface, this method utilizes the spatial centerline of the point cloud as the initial reference boundary (i.e., the source line) to compute the geodesic cost field across the entire surface.

Assuming the center coordinate along the transverse axis is X_{center}, the set of points closest to this central axis is selected as the source point set S. By employing the Dijkstra shortest path algorithm, wavefront propagation expands outward along the anisotropic graph G to determine the minimum accumulated cost from the source set S to all nodes $\mathbf{p}_i$:

$$C_c(\mathbf{p}_i) = \min_{\mathbf{s} \in S} \text{Cost}(\mathbf{s}, \mathbf{p}_i) \tag{3}$$

Consequently, the geodesic distance value C_c for each point within the coarse layer is determined. The detailed execution procedure of this cost field propagation is summarized in Algorithm 1.

The coarse point cloud P_c lacks the sufficient density required to provide a smooth basis for path sampling. Therefore, the computed cost values C_c are mapped back to the dense fine point cloud P_f using the inverse distance weighting (IDW) interpolation algorithm. The IDW mapping process is shown in Fig. 4(a). As illustrated, a thicker connection line between the target point and its neighboring point indicates a shorter spatial distance, which correspondingly translates to a larger interpolation weight.

Specifically, for any target point $\mathbf{q} \in P_f$ in the fine layer, a k-d tree is utilized to identify its k nearest neighbors $\mathcal{N}_q$ within the coarse layer. The smoothed cost value $C_f(\mathbf{q})$ for the target point is calculated as:

$$C_f(\mathbf{q}) = \sum_{j \in \mathcal{N}_q} \hat{\omega}_j C_c(\mathbf{p}_j) \tag{4}$$

Algorithm 1 Dijkstra-based Geodesic cost field generation.

Input: Anisotropic weighted graph $G = (V, E)$, source point set S, edge weights $w_{\mathbf{u},\mathbf{v}}$
Output: Geodesic distance cost field C_c for all $\mathbf{p}_i \in V$

Initialization:
for each node $\mathbf{p}_i \in V$ **do**
  $C_c(\mathbf{p}_i) \leftarrow \infty$
end for
Create an empty priority queue Q
for each source node $\mathbf{s} \in S$ **do**
  $C_c(\mathbf{s}) \leftarrow 0$
  Insert $\mathbf{s}$ into Q with priority $C_c(\mathbf{s})$ ▷ Only push sources initially
end for
Wavefront Propagation: ▷ Expand from sources
while Q is not empty **do**
  $\mathbf{u} \leftarrow$ Extract node with minimum cost from Q
  for each neighbor $\mathbf{v}$ of $\mathbf{u}$ where $(\mathbf{u}, \mathbf{v}) \in E$ **do**
    $alt \leftarrow C_c(\mathbf{u}) + w_{\mathbf{u},\mathbf{v}}$
    if $alt < C_c(\mathbf{v})$ **then**
      $C_c(\mathbf{v}) \leftarrow alt$
      Insert or Update $\mathbf{v}$ in Q with new priority $C_c(\mathbf{v})$
    end if
  end for
end while
return C_c

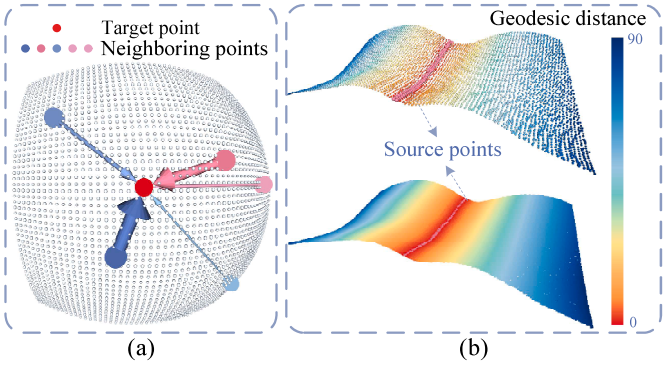


Fig. 4. Spatial mapping and cost field evaluation. (a) Interpolation mapping using Inverse Distance Weighting (IDW); (b) Heatmap distribution of the geodesic distance cost field.

where the normalized weight $\hat{\omega}_j$ is inversely proportional to the square of the spatial distance:

$$\hat{\omega}_j = \frac{\frac{1}{\|\mathbf{q}-\mathbf{p}_j\|_2^2+\epsilon}}{\sum_{i\in\mathcal{N}_q}\frac{1}{\|\mathbf{q}-\mathbf{p}_i\|_2^2+\epsilon}} \tag{5}$$

where ϵ is an infinitesimal constant introduced to prevent division by zero. Ultimately, this dense mapping procedure achieves efficient smoothing and noise reduction for the discrete distance field. Fig. 4(b) illustrates the final geodesic distance-based cost field heatmap.

2.4. Final trajectory extraction and refinement

Based on the continuous cost values C_f within the fine layer, 3D scanning trajectories with equidistant spacing can be generated directly through the extraction of isolines. By defining the effective width of the ultrasound probe as W_{probe} and the desired overlap as W_{overlap}, the theoretical geodesic spacing Δd between adjacent trajectories is calculated as $\Delta d = W_{\text{probe}} - W_{\text{overlap}}$. Consequently, the data point set M_m corresponding to the mth trajectory is formulated as:

$$M_m = \left\{ \mathbf{q} \in P_f \;\middle|\; |C_f(\mathbf{q}) - m \cdot \Delta d| < \delta \right\} \tag{6}$$

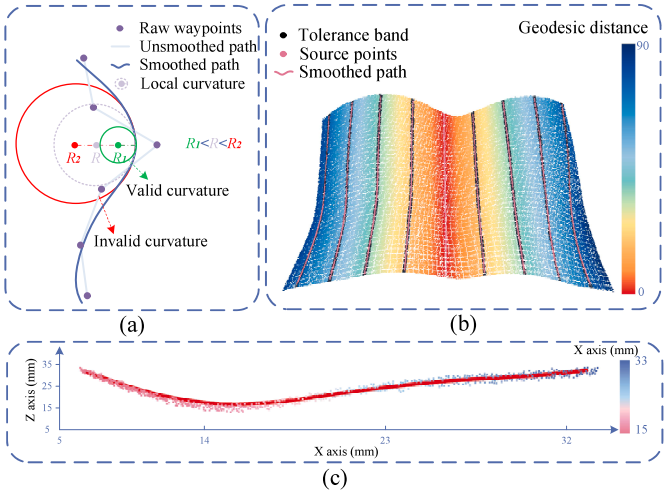


Fig. 5. Isoline extraction and smooth fitting. (a) Maximum curvature constraint; (b) Extracted isolines; (c) Smoothed trajectory via NURBS fitting.

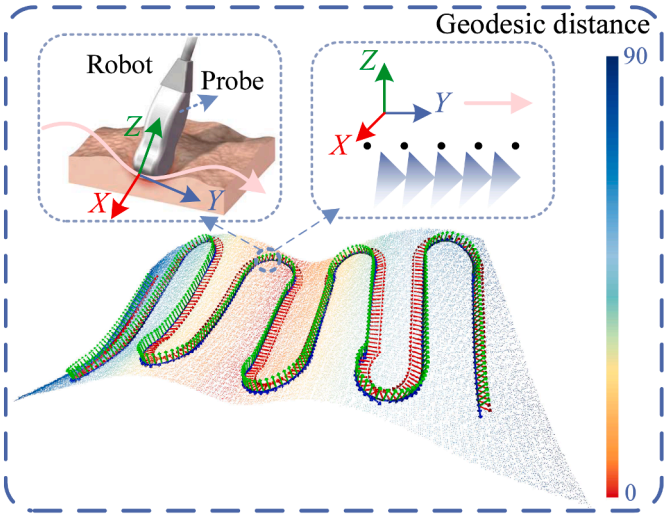


Fig. 6. Final generated trajectories.

where δ denotes the extraction tolerance band.

To mitigate noise within the point set and satisfy the kinematic smoothness requirements of the robotic admittance control system, a cubic non-uniform rational B-spline (NURBS) incorporating a maximum curvature constraint is utilized to parametrically fit M_m (Tan et al., 2022). Let $\mathbf{r}(u)$ denote the parameterized curve, whose curvature $\kappa(u)$ is mathematically defined as:

$$\kappa(u) = \frac{\|\mathbf{r}'(u) \times \mathbf{r}''(u)\|_2}{\|\mathbf{r}'(u)\|_2^3} \tag{7}$$

To handle potential kinematic limitations, the algorithm incorporates an iterative repair strategy. Whenever the local maximum curvature exceeds the threshold (i.e., $\max(\kappa) > 1/R_{\min}$, where $R_{\min}$ represents the minimum safe turning radius of the robot), the spline smoothing parameter s is automatically increased, and local mean filtering is applied to the anomalous points. This procedure is iteratively repeated until the robot's kinematic constraints are strictly satisfied, which ultimately yields the final smooth 3D trajectory. The maximum curvature constraint process is illustrated in Fig. 5(a). Fig. 5(b) shows the result of isoline extraction and path fitting, while the Fig. 5(c) demonstrates the NURBS fitting process for the points within the tolerance band.

To establish a continuous scanning pattern, the individual equidistant trajectories are sequentially linked via a boustrophedon (zig-zag)

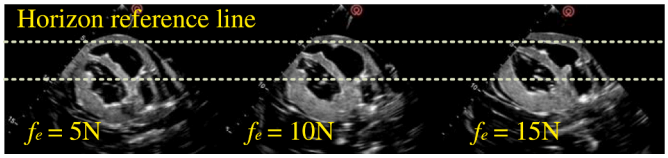


Fig. 7. Ultrasound images of the same tissue under different contact forces.

strategy. Specifically, all trajectories are sorted along the principal spread axis, with the directional order of alternating trajectories reversed. To bridge adjacent endpoints while adhering to human geometric constraints, a surface-adherent U-turn generation method is proposed. An outward-extending semi-circular arc is constructed based on local surface normals, snapped to the dense point cloud via a k-d tree search, and smoothed using spline interpolation. This strategy effectively mitigates kinematic discontinuities, ensuring global C^2 continuity across the entire path.

Upon acquiring the global continuous smooth curve $\mathbf{r}(u)$, discrete waypoints are subsequently sampled along the trajectory using a fixed step size. To execute the robotic scanning task, a local coordinate frame is established at each discrete waypoint to define the spatial pose of the ultrasound probe. Specifically, the outward surface normal vector at the current point is designated as the Z-axis, which aligns with the longitudinal axis of the probe. Furthermore, the forward direction pointing toward the subsequent waypoint defines the initial X-axis vector. To ensure strict spatial orthogonality, this raw forward vector is projected onto the plane perpendicular to the Z-axis to form the true X-axis, and the Y-axis is derived via the cross product of the Z- and X- axes. Ultimately, the local coordinate system constructed at each discrete point completely determines the desired spatial pose of the ultrasound probe during the scanning process. The final generated trajectories are shown in Fig. 6.

3. Disturbance observer-based model predictive admittance control

3.1. Interaction modeling between probe and human tissue

In robotic ultrasound scanning, the probe is typically mounted on the end-effector of a robotic manipulator. As discussed previously, maintaining the probe axis perpendicular to the body surface is critical for image quality. Fig. 7 compares ultrasound images of the same tissue under different contact forces, which reveal that tissue deformation varies significantly with the applied force. To ensure optimal image quality, the probe axis must remain aligned with the surface normal while regulating a constant contact force. Fig. 8 illustrates the interaction model between the ultrasound probe and the human body. To facilitate dynamic analysis, the tissue is modeled as a linear spring:

$$f = \begin{cases} -k_h(x - x_h), & x < x_h \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where k_h denotes the tissue stiffness, x represents the actual interaction position, x_h denotes the position of the body surface, and f is the theoretical interaction force.

For force control, the tissue position and stiffness are treated as time-varying environmental parameters, and the probe-tissue interaction is modeled as a second-order mass-spring-damper system. Classical admittance control (AC) achieves compliant force regulation by adjusting position commands in response to interaction forces. For an n-degree-of-freedom manipulator operating in Cartesian space, the standard admittance law is defined as:

$$M(\ddot{X}_c - \ddot{X}_r) + B(\dot{X}_c - \dot{X}_r) + K(X_c - X_r) = F - F_t \tag{9}$$

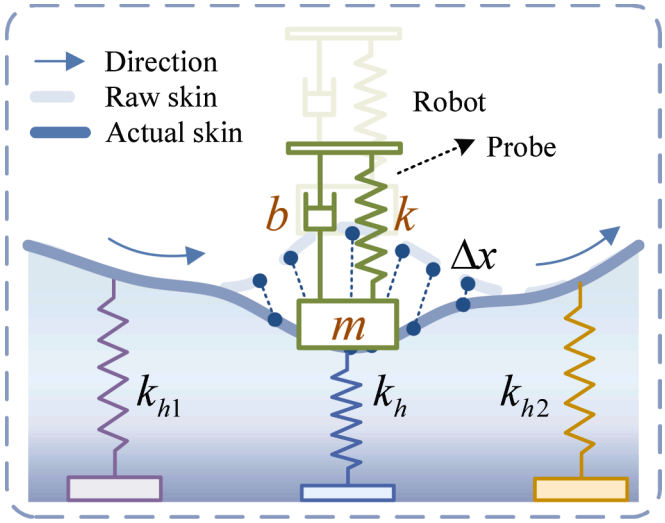


Fig. 8. Interaction model between the ultrasound probe and the human body.

where $M, B, K \in \mathbb{R}^{n\times n}$ are the symmetric positive-definite virtual mass, damping, and stiffness matrices, respectively. Vectors $X_r, X_c \in \mathbb{R}^n$ denote the reference and commanded trajectories, while $F, F_t \in \mathbb{R}^n$ represent the actual and desired interaction forces.

Given that the probe-tissue interaction can be decoupled across Cartesian axes, the admittance model is simplified to a one-dimensional form:

$$m(\ddot{x}_c - \ddot{x}_r) + b(\dot{x}_c - \dot{x}_r) + k(x_c - x_r) = e_f \tag{10}$$

where $e_f = f - f_t$, and m, b, k represent the virtual mass, damping, and stiffness coefficients. Here, f and f_t correspond to the measured and desired interaction forces along the surface normal.

Assuming the robot provides high-precision position tracking, we consider the tracking error $x - x_c$ to be negligible. Any unmodeled dynamics and residual errors are encapsulated into a lumped disturbance, which is subsequently compensated for within the DO-MPAC framework.

Remark 1. Based on the analysis in Section 2, ultrasound scanning is a process that considers probe pose constraints. Section 1 notes that ultrasound image quality primarily depends on normal contact, while tangential motion provides spatial coverage. Therefore, this paper models admittance control only in the normal direction. Additionally, improved imaging reproducibility is a key advantage of robotic ultrasound scanning. Simplifying the system to a 1D normal admittance control allows it to reproduce images at the same location quickly and reliably. This requires only normal adjustments and avoids complex tuning in other directions. Similar 6-DOF to 1D simplifications appear in recent robotic ultrasound research (Jiang et al., 2024; Li et al., 2026a,b; Tan et al., 2022). During continuous scanning on curved surfaces, tangential friction, orientation errors, and multi-axis coupling cause cross-axis disturbances to the normal force. Instead of computing a complex 6-DOF dynamic model, the proposed framework groups these unmodeled effects into the lumped disturbance term $D(t)$ in Eq. (19). The designed ESO in Eq. (20) then estimates and compensates for these multi-axis impacts. This approach keeps the model mathematically concise and preserves scanning reproducibility. Note that this simplification assumes the robot provides ideal tangential and orientation tracking, and that the cross-axis disturbances remain bounded.

3.2. System formulation

Following the analysis of the admittance model, desired force tracking can be achieved by setting $k = 0$. It is crucial to clarify that k represents the software-defined virtual stiffness in the admittance controller,

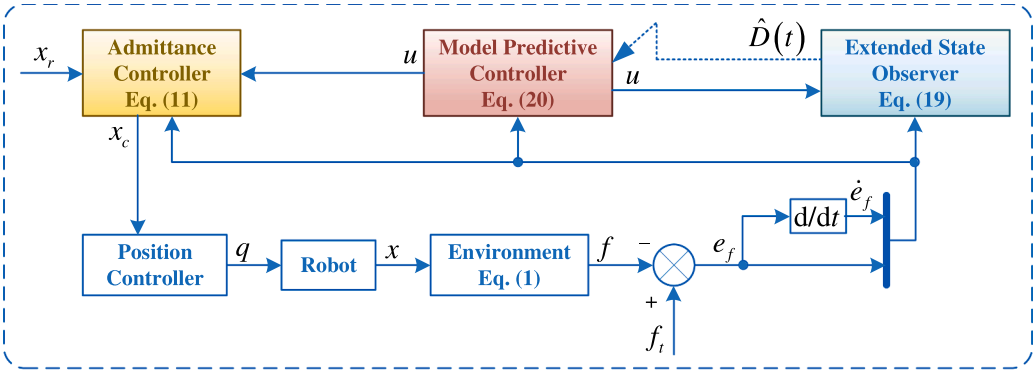


Fig. 9. Schematic of the proposed DO-MPAC strategy.

rather than the physical stiffness of the robotic joints or the human tissue (k_h). During steady-state contact where velocity and acceleration approach zero ($\ddot{x}_c = \dot{x}_c = 0$), the original admittance model (Eq. (10)) degenerates to $e_f = k(x_c - x_r)$. To compress the tissue and apply a constant force, the probe must physically deviate from its reference trajectory ($x_c \neq x_r$). Consequently, a non-zero virtual stiffness ($k > 0$) induces a virtual restoring force that pulls the end-effector toward x_r, inherently causing a steady-state tracking error. Setting $k = 0$ mathematically eliminates this virtual spring effect, allowing the manipulator to exhibit force-guided compliance and achieve nearly zero steady-state error. This configuration is effective for force tracking tasks and has been validated in Jung et al. (2004). Beyond the elimination of steady-state errors, setting $k = 0$ is structurally imperative for the downstream DO-MPAC formulation, as clarified in Remark 2. Therefore, the admittance model simplifies to:

$$m(\ddot{x}_c - \ddot{x}_r) + b(\dot{x}_c - \dot{x}_r) = e_f \tag{11}$$

To integrate model predictive control (MPC) into the AC framework, we introduce a virtual input u, which acts as an auxiliary control variable. Let $\Delta x = x_c - x_r$. The admittance model is reformulated as:

$$m\Delta\ddot{x} + b\Delta\dot{x} = e_f + u \tag{12}$$

By rearranging the terms, the commanded acceleration can be derived as:

$$\ddot{x}_c = \ddot{x}_r + \frac{1}{m}(e_f + u - b(\dot{x}_c - \dot{x}_r)) \tag{13}$$

Given the force error $e_f = f - f_t = -k_h(x_c - x_h) - f_t$, its first and second-order derivatives with respect to time are expressed as:

$$\dot{e}_f = -\dot{k}_h(x_c - x_h) - k_h(\dot{x}_c - \dot{x}_h) \tag{14}$$

$$\ddot{e}_f = -k_h\ddot{x}_c + k_h\ddot{x}_h - 2\dot{k}_h(\dot{x}_c - \dot{x}_h) - \ddot{k}_h(x_c - x_h) \tag{15}$$

Substituting Eq. (13) into Eq. (15), the dynamics of the force error are obtained as:

$$\begin{aligned}\ddot{e}_f = &- k_h\left[\ddot{x}_r + \frac{1}{m}(e_f + u - b(\dot{x}_c - \dot{x}_r))\right] \\ &+ k_h\ddot{x}_h - 2\dot{k}_h(\dot{x}_c - \dot{x}_h) - \ddot{k}_h(x_c - x_h)\end{aligned} \tag{16}$$

To account for the time-varying environment stiffness and separate the nominal model from uncertainties, we represent k_h as $k_h = k_{h0} + \Delta k_h$, where k_{h0} is the nominal stiffness and Δk_h denotes the stiffness fluctuation. Substituting this into Eq. (13) yields:

$$\ddot{e}_f = M_0 u + H_n + D(t) \tag{17}$$

where $M_0 = -k_{h0}/m$ and $H_n = -\frac{k_{h0}}{m}e_f + \frac{k_{h0}b}{m}(\dot{x}_c - \dot{x}_r) - k_{h0}\ddot{x}_r$. The lumped disturbance term $D(t)$ is defined as:

$$\begin{aligned}D(t) = &- \frac{\Delta k_h}{m}(u + e_f - b(\dot{x}_c - \dot{x}_r)) \\ &- \Delta k_h\ddot{x}_r + k_h\ddot{x}_h - 2\dot{k}_h(\dot{x}_c - \dot{x}_h) - \ddot{k}_h(x_c - x_h)\end{aligned} \tag{18}$$

Notably, while Eq. (18) formulates $D(t)$ from environmental dynamics, it practically encapsulates low-frequency unmodeled dynamics like friction and modeling errors for ESO compensation. Meanwhile, high-frequency uncertainties like sensor noise are effectively mitigated by the virtual mass m and the low-pass filtering of the ESO bandwidth ω_o.

3.3. DO-MPAC design

For the second-order system (17), a third-order extended state observer (ESO) is introduced to monitor the unknown disturbance $D(t)$ in real time. The state space is defined as $l_1 = e_f$, $l_2 = \dot{e}_f$, and $l_3 = D(t)$. The continuous-time state-space equations for the true force tracking error are given by:

$$\begin{cases}\dot{l}_1 = l_2 \\ \dot{l}_2 = l_3 + M_0 u + H_n \\ \dot{l}_3 = \dot{D}(t)\end{cases} \tag{19}$$

To observe these states in real time, a third-order ESO is designed. Let the observer state variables be z_1, z_2, z_3, and define the observation error as $e_o = z_1 - l_1$. The ESO dynamics are then constructed as:

$$\begin{cases}\dot{z}_1 = z_2 - \beta_1 e_o \\ \dot{z}_2 = z_3 - \beta_2 e_o + M_0 u + H_n \\ \dot{z}_3 = -\beta_3 e_o\end{cases} \tag{20}$$

where the observer gains are selected as $\beta_1 = 3\omega_o$, $\beta_2 = 3\omega_o^2$, and $\beta_3 = \omega_o^3$. Define the state estimation error vector as $\boldsymbol{e} = [e_1, e_2, e_3]^T$, where the components are $e_1 = z_1 - l_1$ (i.e., e_o), $e_2 = z_2 - l_2$, and $e_3 = z_3 - l_3$. Subtracting the actual system equations from the observer equations yields the ESO error dynamics:

$$\begin{cases}\dot{e}_1 = \dot{z}_1 - \dot{l}_1 = e_2 - \beta_1 e_1 \\ \dot{e}_2 = \dot{z}_2 - \dot{l}_2 = e_3 - \beta_2 e_1 \\ \dot{e}_3 = \dot{z}_3 - \dot{l}_3 = -\beta_3 e_1 - \dot{D}(t)\end{cases} \tag{21}$$

The actual control input u is designed as:

$$u = \frac{u_0 - H_n - z_3}{M_0} \tag{22}$$

where u_0 represents the virtual control law to be designed. Substituting u back into the error dynamics (17) yields:

$$\ddot{e}_f = u_0 - z_3 + l_3 \approx u_0 \tag{23}$$

To systematically design the virtual control law u_0, an MPC scheme is introduced. Let the state vector be $\mathbf{X} = [e_f, \dot{e}_f]^T$. Applying forward Euler discretization with a sampling time T_s to the system $\ddot{e}_f = u_0$, we obtain:

$$\mathbf{X}(k+1) = A\mathbf{X}(k) + Bu_0(k) \tag{24}$$

where $A = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} T_s^2/2 \\ T_s \end{bmatrix}$. The system output is the desired force error, defined as $E_f(k) = C\mathbf{X}(k) = e_f(k)$, with $C = [1, 0]$. Over a prediction horizon N_p, the control objective is to drive the future force error e_f toward zero while preventing abrupt variations in the control action u_0. Accordingly, the cost function J is formulated as:

$$J = \min_{\mathbf{U}_0} \sum_{i=1}^{N_p} \left\| e_f(k+i) \right\|_Q^2 + \sum_{i=0}^{N_p-1} \left\| u_0(k+i) \right\|_R^2 \tag{25}$$

The predicted outputs over the future N_p steps are aggregated into a vector:

$$\mathbf{E}_{fp} = \Psi \mathbf{X}(k) + \Theta \mathbf{U}_0 \tag{26}$$

where Ψ and Θ represent the standard prediction matrices. The analytical solution for the optimal control sequence is derived by substituting this prediction model into the cost function and setting its derivative with respect to $\mathbf{U}_0$ to zero:

$$\mathbf{U}_0^* = -(\Theta^T Q \Theta + R)^{-1} \Theta^T Q \Psi \mathbf{X}(k) \tag{27}$$

The first element of this optimal sequence is extracted as the current control command, denoted as $u_0(0)$. Substituting this into the input transformation yields:

$$u = \frac{u_0(0) - H_n - z_3}{M_0} \tag{28}$$

By pre-computing the time-invariant gain matrix in Eq. (27) offline, the online DO-MPAC only requires fixed-size matrix-vector multiplications, achieving effectively $O(1)$ complexity with respect to the prediction horizon N_p. Finally, the control command is obtained by substituting the above expression (28) into the admittance model (13):

$$\ddot{x}_c = \ddot{x}_r + \frac{1}{m}\left[e_f + \left(\frac{u_0(0) - H_n - z_3}{M_0} \right) - b(\dot{x}_c - \dot{x}_r) \right] \tag{29}$$

Fig. 9 depicts the block diagram for the developed DO-MPAC framework.

3.4. Stability and boundedness analysis of DO-MPAC

Assumption 1. The unknown lumped disturbance $D(t)$ is continuously differentiable, and its rate of change is bounded by a finite positive constant L_D, such that $|\dot{D}(t)| \le L_D$.

Theorem 1. *For the admittance system (11)-(29), if Assumption 1 holds and the finite-horizon MPC parameters (N_p, Q, and R) are selected such that the nominal closed-loop system is Schur-stable, the closed-loop tracking states $\mathbf{X}(k) = [e_f(k), \dot{e}_f(k)]^T$ are uniformly ultimately bounded (UUB).*

Proof. Part A. Stability Analysis of the ESO

Rewriting Eq. (21) in a compact matrix form yields:

$$\dot{\boldsymbol{e}} = A_e \boldsymbol{e} + E_e \dot{D}(t) \tag{30}$$

where the error system matrix A_e and the disturbance input matrix E_e are defined as:

$$A_e = \begin{bmatrix} -\beta_1 & 1 & 0 \\ -\beta_2 & 0 & 1 \\ -\beta_3 & 0 & 0 \end{bmatrix}, \quad E_e = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}. \tag{31}$$

According to the analysis in active disturbance rejection control (Gao et al., 2003), by applying the bandwidth parameterization method to configure the observer gains as $\beta_1 = 3\omega_o$, $\beta_2 = 3\omega_o^2$, and $\beta_3 = \omega_o^3$ ($\omega_o > 0$), all eigenvalues of matrix A_e are placed at $-\omega_o$ in the complex plane. This configuration ensures that A_e is a strictly Hurwitz matrix.

By defining $h = -\dot{D}(t)$, the error dynamics in Eq. (30) become $\dot{\boldsymbol{e}} = A_e \boldsymbol{e} - E_e h$. Given that A_e is strictly Hurwitz and h is bounded by L_D (Assumption 1), the system achieves bounded-input bounded-state (BIBS) stability (Gao et al., 2003). Therefore, the estimation error $\tilde{D}(t) = z_3(t) - l_3(t)$ is ultimately bounded by a positive constant μ (i.e., $\limsup_{t\to\infty} |\tilde{D}(t)| \le \mu$).

Part B. Stability Analysis of the MPC

While the DO-MPAC formulation (e.g., Eq. (24)) relies on a nominal, disturbance-free model for optimization, evaluating the physical closed-loop stability requires accounting for the uncompensated ESO residual, $\tilde{D}(t) = z_3(t) - l_3(t)$. Substituting the actual control law (22) into the physical force error dynamics yields:

$$\ddot{e}_f = u_0 - \tilde{D}(t) \tag{32}$$

Discretizing this equation produces the perturbed state-space model:

$$\mathbf{X}(k+1) = A\mathbf{X}(k) + Bu_0(k) + W(k) \tag{33}$$

where matrices A and B match the nominal prediction model Eq. (24), and $W(k) = [-\frac{T_s^2}{2}\tilde{D}(k), -T_s\tilde{D}(k)]^T$ is the discretized perturbation induced by $\tilde{D}(t)$. Since $\limsup_{t\to\infty} |\tilde{D}(t)| \le \mu$ (**Part A**), $W(k)$ is strictly bounded by a positive constant $w_{\max}$ (i.e., $\|W(k)\| \le w_{\max}$).

The unconstrained MPC law (27) operates as a nominal linear state feedback $u_0(k) = -K_{mpc}\mathbf{X}(k)$, where K_{mpc} corresponds to the first row of $(\Theta^T Q\Theta + R)^{-1}\Theta^T Q\Psi$. For the adopted finite-horizon MPC, a sufficiently long prediction horizon N_p and positive-definite weighting matrices ($Q > 0$, $R > 0$) are selected to obtain a stabilizing feedback gain K_{mpc} for the nominal system (Mayne et al., 2000; Rawlings & Mayne, 2009). Consequently, the nominal closed-loop matrix $A_K = A - BK_{mpc}$ is Schur-stable, which is verified by checking that all eigenvalues lie strictly inside the unit circle.

Applying this nominal control law to the perturbed physical system yields the actual closed-loop dynamics:

$$\mathbf{X}(k+1) = A_K\mathbf{X}(k) + W(k) \tag{34}$$

This formulation reveals the inherent robustness of the controller. Despite the continuous external perturbation $W(k)$, the strictly Schur-stable matrix A_K effectively suppresses the disturbance. According to discrete-time input-to-state stability (ISS) theory (Jiang & Wang, 2001; Khalil, 2002), applying a bounded input to an asymptotically stable nominal system naturally preserves ISS. Consequently, $\limsup_{k\to\infty} \|\mathbf{X}(k)\| \le c_w w_{\max}$, where $c_w > 0$ is a constant related to A_K. This proves that the closed-loop DO-MPAC system is uniformly ultimately bounded (UUB). □

Remark 2. Setting the virtual stiffness to zero prevents coupling between force and position errors in Eq. (16). This simplifies the MPC design by maintaining a compact 2D state vector $\mathbf{X} = [e_f, \dot{e}_f]^T$ instead of a 4D space, reducing computational load and avoiding a multi-objective cost function. In practice, obtaining the state derivative $\dot{e}_f$ via direct numerical differentiation amplifies sensor noise. Therefore, the physical hardware measures only the raw force error e_f (i.e., l_1). The ESO then reconstructs the remaining dynamics: z_2 provides $\dot{e}_f$, and z_3 estimates higher-order coupling effects. This observer-based approach supplies stable states for the MPC scheme in Eq. (24) without numerical derivation.

Remark 3. As shown in Eq. (16), the dynamics of the force error, e_f, in traditional AC are coupled with the virtual mass m and damping b. Consequently, tuning these parameters requires a trade-off between tracking agility and physical compliance. The proposed DO-MPAC framework suppresses this coupling through feedback linearization. By integrating the nominal dynamics, H_n, into the inverse dynamics control law, a feedforward control term is introduced to compensate for the dynamic effects of m and b. As established in (22), with accurate disturbance estimation ($z_3 \approx D(t)$), the closed-loop force error dynamics are nominally transformed into a standard double integrator (i.e., $\ddot{e}_f \approx u_0$). Thus, the system's transient convergence rate is governed primarily by the MPC weighting matrices (Q, R). However, it is crucial to note that this simplification depends on the ESO's tracking bandwidth. Highly transient events, such as sudden environmental stiffness jumps, can cause the ESO to briefly lag behind the actual disturbance ($z_3 \neq D(t)$). During these brief estimation delays, uncompensated residual forces act directly on the physical admittance model. Therefore, rather than rendering m and b

obsolete, DO-MPAC fundamentally separates their functional roles. The control law dictates the nominal tracking agility, while m and b are tuned to absorb high-frequency impacts. This mechanism suppresses the traditional parameter coupling, simplifies the tuning process, and enhances robustness against complex environmental variations.

Remark 4. It is worth noting that the MPC formulation in Eq. (25) omits explicit state and input constraints primarily to preserve the analytical solution shown in Eq. (27). While explicitly constrained MPC strictly bounds system variables, online numerical quadratic programming (QP) solving increases the computational burden and compromises the determinism of execution times in high-frequency robotic control. Alternatively, this paper adopts a hierarchical safety strategy. At the optimization level, the objective function provides a penalty-based soft constraint mechanism. The weighting matrices Q and R penalize force tracking errors and control variations, respectively, thereby suppressing excessive force fluctuations and aggressive control actions. At the execution level, hard saturation mechanisms are introduced to enhance interaction safety. Specifically, strict boundaries are imposed on the control input u_0 and the admittance-induced positional deviation Δx. According to Eqs. (23) and (29), these constraints physically limit the transient control behavior ($\ddot{x}_c$) and probe motion, thereby reducing the risk of excessive mechanical interaction under anomalous conditions. The proposed framework does not preclude explicit MPC constraints. The explicit input or state constraints can be embedded directly into the MPC optimization problem. However, this will transform the analytical optimization into a numerical constrained optimization problem.

4. Simulation studies

4.1. Simulation setup

In this section, several simulations are conducted in MATLAB (2024b) to evaluate the performance of the proposed DO-MPAC method. To comprehensively assess the tracking accuracy and robustness of the proposed framework, several representative admittance controllers are selected for comparative analysis. These baselines include the conventional admittance controller (AC) in Tan et al. (2022), the adaptive admittance controller (AAC) in Jiang et al. (2024), and the disturbance observer-based sliding mode admittance controller (DOBSMAC) in Li et al. (2026a).

As the foundational baseline, the conventional AC scheme adopted in Tan et al. (2022) is expressed as Eq. (11). Furthermore, the AAC method proposed in Jiang et al. (2024) is formulated as follows:

$$\begin{cases} \hat{k}_h = \hat{k}_h(0) + \alpha \int_0^t x_c(\hat{f} - f)dt \\ \hat{x}_h = \hat{x}_h(0) - \beta \int_0^t \frac{\hat{f}-f}{\hat{k}_h}\left(\frac{\alpha}{\beta} x_c \hat{x}_h + 1\right)dt \\ \hat{f}(t) = \hat{k}_h(\hat{x}_h - x_c) \end{cases} \tag{35}$$

where $\hat{f}$ represents the estimated contact force, and $\hat{k}_h$ and $\hat{x}_h$ denote the estimated environment stiffness and position, respectively. Finally, α and β act as positive scalar constants to tune the adaptive laws.

To provide a benchmark for robust nonlinear control, the DOBSMAC strategy in Li et al. (2026a) is configured as:

$$\begin{cases} s = Ce_f + \dot{e}_f \\ u = \frac{1}{G_0}\left(-k_s s - \eta \mathrm{sat}(s) - \hat{d} - h_{nom}\right) \\ \dot{\hat{s}} = G_0 u + h_{nom} + \hat{d} + L_1(s - \hat{s}) \\ \dot{\hat{d}} = L_2(s - \hat{s}) \end{cases} \tag{36}$$

where s defines the first-order sliding manifold, which is parameterized by a positive constant C. The virtual control input u is synthesized to enforce the reaching condition of the sliding surface, incorporating k_s as the linear feedback gain and η as the robust switching gain. In this formulation, G_0 and h_{nom} represent the nominal mass and the nominal intrinsic dynamics of the admittance model, respectively. Meanwhile, an extended state observer structure reconstructs the lumped disturbance $\hat{d}$ and the sliding variable dynamics $\hat{s}$ in real time, with L_1 and L_2 serving as the positive observer gains.

Control efficacy is quantitatively assessed using six standard indices. First, the rise time (T_{rise}) evaluates transient agility, defined as the duration from initial contact t_c to first reaching the target force f_t:

$$T_{rise} = \inf\{t > t_c \mid f(t) \geq f_t\} - t_c \tag{37}$$

where $t_a = t_c + T_{rise}$ denotes the arrival instant. Second, the settling time (T_s) measures convergence speed, representing the time required to continuously remain within a ±5% error band over the total duration t_r:

$$T_s = \inf\{t > t_c \mid \forall \tau \in [t, t_r], |f(\tau) - f_t| \leq 0.05 f_t\} - t_c \tag{38}$$

Third, the maximum force deviation (MFD) characterizes the maximum absolute transient deviation percentage from the target force after t_a:

$$\mathrm{MFD} = \sup_{t \in [t_a, t_r]} \left| \frac{f(t) - f_t}{f_t} \right| \times 100\% \tag{39}$$

Fourth, steady-state tracking accuracy is evaluated by the integral of absolute error (IAE) and time-weighted absolute error (ITAE) (Yi & Zhai, 2019):

$$\mathrm{IAE} = \int_0^{t_r} |e_f(t)| \mathrm{d}t, \quad \mathrm{ITAE} = \int_0^{t_r} t \cdot |e_f(t)| \mathrm{d}t \tag{40}$$

Finally, to quantify high-frequency chattering, the high-frequency energy (HFE) measures oscillation energy beyond a cutoff frequency f_c, where $P(f)$ is the force amplitude spectrum and $f_{\max}$ is the maximum frequency. In this study, the frequency evaluation range is specifically set from $f_c = 20$ Hz to $f_{\max} = 500$ Hz:

$$\mathrm{HFE} = \int_{f_c}^{f_{max}} |P(f)|^2 \mathrm{d}f \tag{41}$$

Within the simulation framework, the environmental parameters are modeled to reflect the realistic physiological characteristics of the human surfaces. The time-varying tissue stiffness $k_h(t)$ and respiratory-induced surface motion $x_h(t)$ are configured as follows:

$$\begin{cases} k_h(t) = k_{h0} + \Delta k_h \sin(2\pi f_k t) \\ x_h(t) = A_r \sin(2\pi f_r t) \end{cases} \tag{42}$$

where $k_{h0} = 2500$ N/m denotes the nominal stiffness of the muscular tissue. Δk_h and f_k represent the amplitude and frequency of the spatial stiffness variation as the probe scans across different anatomical tissues. Furthermore, A_r signifies the amplitude of human respiration, and f_r denotes the respiratory frequency.

An evaluation of the controller performance is achieved by establishing two distinct sets of environmental parameters. The first set, denoted as **Case 1**, simulates a relatively stable environment with minor physiological perturbations, where $\Delta k_h = 0.3$ N/m, $f_k = 0.1$ Hz, $A_r = 0.0002$ m, and $f_r = 0.1$ Hz. Conversely, the second set, denoted as **Case 2**, represents a challenging scenario with significant respiratory motion and severe stiffness fluctuations, characterized by $\Delta k_h = 1200$ N/m, $f_k = 0.3$ Hz, $A_r = 0.005$ m, and $f_r = 0.3$ Hz.

The admittance parameters are selected to satisfy physical hardware constraints and clinical safety requirements. First, considering the actual 2.7 kg end-effector payload (ultrasound probe, camera, force sensor, and connectors), the virtual mass is set to $m = 3.5$ kg. This added mass acts as an inertia buffer to filter high-frequency sensor noise and compensate for unmodeled trailing cable dynamics during motion. Second, to determine the optimal virtual damping b, a parameter sweep is conducted using traditional AC under the relatively stable **Case 1**. As shown in Fig. 10, the sweep results reveal a critical trade-off between tracking agility (T_S) and patient safety (maximum transient velocity). Lower damping values accelerate the force response but induce hazardous transient velocities that exceed the strict 0.03 m/s safety threshold during sudden contact. Therefore, a highly conservative damping of $b = 1000$ N·s/m is selected. This damping restricts the maximum speed to approximately 0.01 m/s to guarantee clinical safety, and provides a stable T_s of approximately 1.2 s.

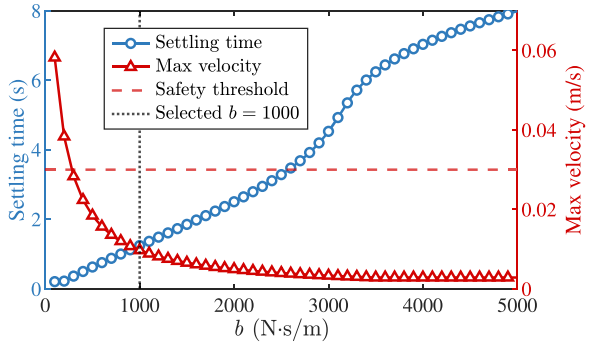


Fig. 10. Parameter sweep analysis of virtual damping b in **Case 1**.

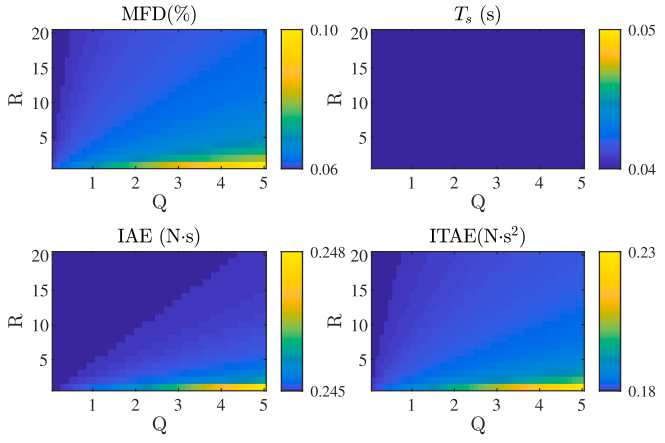


Fig. 11. Parameter sweep analysis of Q, R in **Case 1**.

To ensure a fair comparison, identical admittance parameters ($m = 3.5$, $b = 1000$, and $k = 0$) are applied to all four controllers. The simulation step size is uniformly set to $T = 0.002$ s.

The parameter tuning procedure for the DO-MPAC is outlined as follows. First, the stiffness parameter, k_{h0}, is chosen as 1500 to approximate the surface stiffness of the human body, and the prediction horizon, N_p, is chosen to be 15. As discussed in Remark 3, the ESO bandwidth primarily influences the controller's performance during the initial tracking phase, owing to the large initial estimation error of $D(t)$. Once the ESO accurately tracks $D(t)$, the system performance is predominantly governed by the MPC weighting parameters. Therefore, with ω_o initially set to 100, a parameter sweep is conducted for Q and R in **Case 1**, and their optimal values are determined by the corresponding performance metrics. The results of this parameter sweep are presented in Fig. 11. Based on a comprehensive evaluation of the four metrics, the parameters are set to $Q = 0.5$ and $R = 10$.

Finally, the bandwidth parameter, ω_o, remains to be optimized. The selection operates within a theoretical interval $[\omega_{\min}, \omega_{\max}]$. The lower bound $\omega_{\min}$ should exceed the dominant frequency of environmental disturbances, and the upper bound $\omega_{\max}$ is constrained by the sampling rate and sensor noise limits to avoid resonance. As illustrated in Fig. 12, the parameter sweep conducted for ω_o in **Case 1** demonstrates the impact of bandwidth size on controller performance, providing guidance for actual deployment. Based on the evaluation metrics within this bounded region, ω_o is set to 90 for the simulations.

The specific parameter configurations for the remaining control variables corresponding to each strategy are detailed in Table 1. To ensure a fair comparison, the parameters of each controller are based on their respective literature and fine-tuned for optimal performance in **Case 1**.

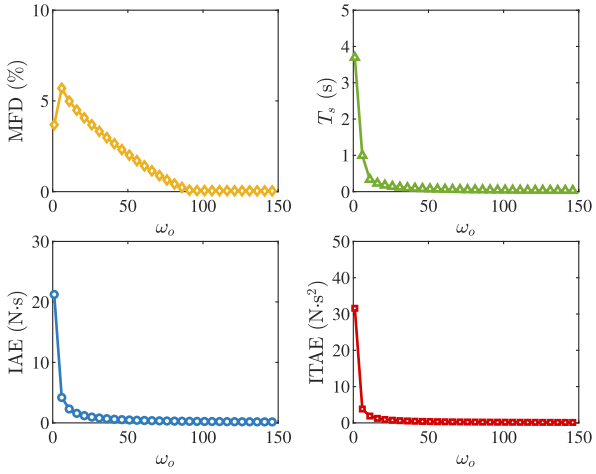


Fig. 12. Parameter sweep analysis of ω_o in **Case 1**.

Table 1
Control parameter configurations for the four controllers in simulations.

Controller	Parameter Settings
AC	$m = 3.5$, $b = 1000$, $k = 0$
AAC	$\alpha = 0.01$, $\beta = 0.01$
DOBSMAC	$C = 20$, $k_s = 150$, $\eta = 10$, $L_1 = 60$, $L_2 = 1200$
DO-MPAC	$k_{h0} = 1500$, $N_p = 15$, $Q = 0.5$, $R = 10$, $\omega_o = 90$

Table 2
Quantitative simulation performance metrics of the evaluated controllers.

Condition	Method	MFD (%)	IAE (N·s)	ITAE (N·s²)
Case 1	AC	1.1698	4.7891	5.1306
	AAC	1.5448	6.5838	7.9286
	DOBSMAC	6.9831	0.6490	0.5815
	DO-MPAC	0.1066	0.2716	0.2038
Case 2	AC	24.2631	7.5905	35.8323
	AAC	25.6664	14.6170	71.0296
	DOBSMAC	10.3077	0.7345	2.4451
	DO-MPAC	0.1712	0.5694	0.5142

4.2. Simulation results

In this subsection, the tracking performance of the proposed DO-MPAC framework is compared with that of three representative admittance controllers. To ensure a fair comparison, all evaluated strategies are tested under identical initial conditions and algorithmic parameters. The quantitative performance metrics are summarized in Table 2, and the response trajectories are plotted in Figs. 13–15 to illustrate the transient and steady-state behaviors.

The force tracking trajectories generated by the four control strategies under **Case 1** and **2** are depicted in Figs. 13 and 14, respectively. In **Case 1**, all evaluated controllers successfully achieve the target force. Notably, DOBSMAC exhibits the most rapid transient response, but it incurs a noticeable overshoot along with minor force fluctuations during the steady-state phase. Although the AC and AAC strategies yield almost zero overshoot, they suffer from the slowest convergence, and their steady-state oscillations are more pronounced than those of DOBSMAC. By contrast, the proposed DO-MPAC delivers a tracking speed second only to DOBSMAC, yet it maintains a minimal overshoot and achieves optimal steady-state performance. Under the extreme conditions of **Case 2**, the superiority of the steady-state behavior of DO-MPAC becomes more evident. Both AC and AAC generate substantial overshoots and severe periodic oscillations. Specifically, AAC experiences the largest fluctuations, which is fundamentally attributed to its high

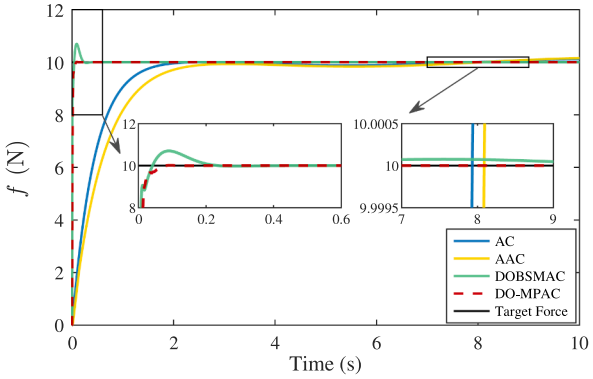


Fig. 13. Comparison of force tracking results in **Case 1**.

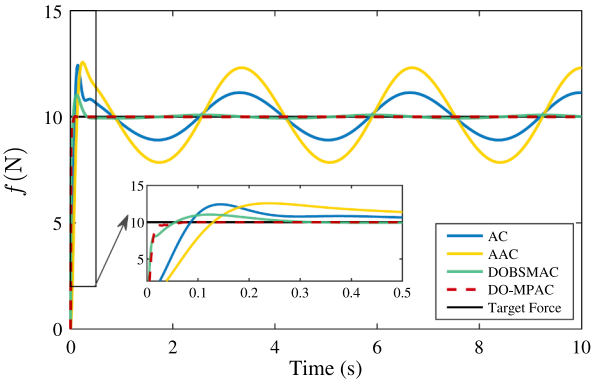


Fig. 14. Comparison of force tracking results in **Case 2**.

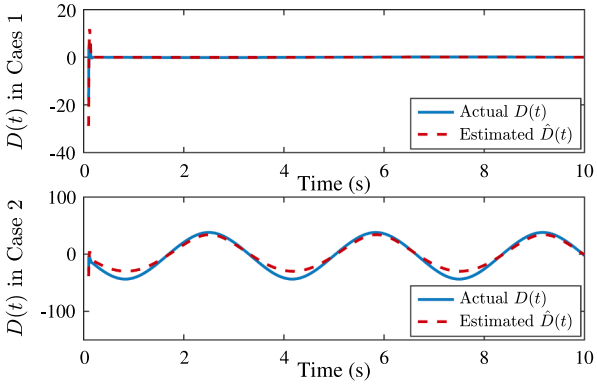


Fig. 15. Disturbance estimation results of the designed ESO.

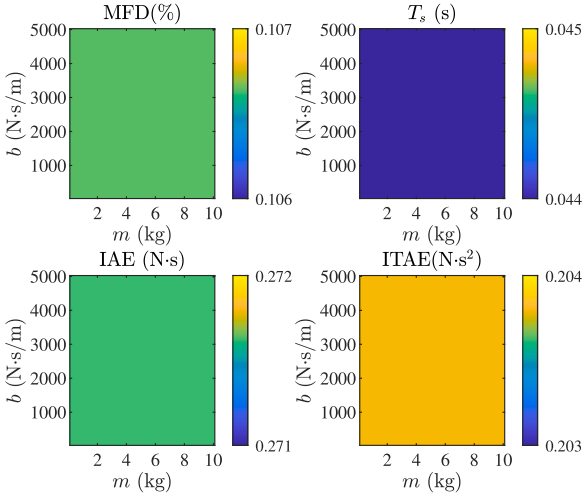


Fig. 16. Robustness of the DO-MPAC against variations in m and b.

sensitivity to environmental parameter variations. Concurrently, while DOBSMAC maintains the fastest response speed, it induces an overshoot significantly larger than that of DO-MPAC. Even though the rise time of DO-MPAC is slightly longer than that of DOBSMAC, it effectively suppresses the large overshoot and continuously preserves stable steady-state tracking capability. Additionally, the disturbance estimation results of the designed ESO in **Cases 1** and **2** are illustrated in Fig. 15, validating its effectiveness.

A quantitative analysis of the metrics in Table 2 further confirms these observations. Compared with the AC, AAC, and DOBSMAC strategies, the proposed DO-MPAC yields absolute reductions in the MFD by 24.09%, 25.50%, and 10.14% in **Case 2**, respectively. Concurrently, the IAE is decreased by absolute margins of 7.02 N · s, 14.05 N · s, and 0.17 N · s, which collectively demonstrates the robust tracking performance of the developed framework.

To verify the reduced dependency of the DO-MPAC's performance on the parameters m and b, as analyzed in Remark 3, a parameter sweep for m and b is conducted in **Case 2**. As illustrated in Fig. 16, the four primary metrics exhibit minimal fluctuations across the variations in m and b. These results confirm that the DO-MPAC effectively reduces the performance dependency on these admittance parameters.

Fundamentally, the aforementioned performance disparities can be attributed to the intrinsic regulatory mechanisms of the respective controllers. Although the conventional AC can accommodate specific constant-stiffness environments, its fixed admittance coefficients limit its adaptability to time-varying environments, which ultimately leads to reduced steady-state accuracy. To address this, the AAC strategy attempts to improve precision by adaptively tuning parameters based on the environment. However, its update law is inherently reactive, relying primarily on past error accumulation. Such a reactive mechanism introduces a noticeable phase lag, which makes the system highly susceptible to larger overshoots and periodic oscillations. Furthermore, by utilizing the state estimation from the disturbance observer combined with the enforced convergence properties of the sliding manifold, DOBSMAC achieves a rapid force-tracking response. Nevertheless, its performance is constrained by the inherent limitations of conventional sliding mode methodologies. Specifically, the employment of a linear sliding surface restricts the dynamic flexibility during the reaching phase, while the discontinuous switching control law inevitably triggers the chattering phenomenon, which degrades the actual interaction quality. In contrast, the proposed DO-MPAC not only effectively suppresses the uncertainties arising from environmental fluctuations via ESO-based disturbance compensation, but also leverages the receding horizon optimization mechanism of MPC. This forward-looking strategy predicts system states and optimizes future control actions during the receding horizon process, which yields optimal contact force transitions. Consequently, it achieves an optimized trade-off between a fast transient response and steady-state stability.

5. Experimental validation

In this section, physical experiments are conducted on a real robotic platform to evaluate the effectiveness of the proposed ultrasound scanning workflow.

5.1. Experimental platform

The experimental platform utilized in this study is independently developed and illustrated in Fig. 17. Table 3 lists the detailed specifications

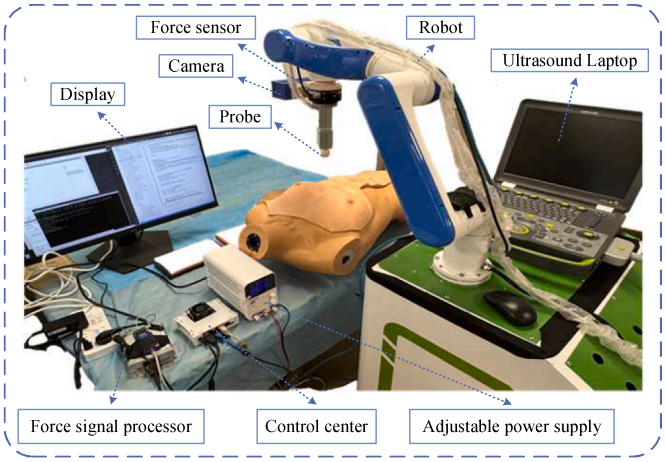


Fig. 17. Robotic ultrasound experimental platform.

Table 3
Detailed specifications of the hardware components.

Component Category	Model / Specification
Collaborative Manipulator	Ti5 Robot Eblm-3 (6-DoF)
Depth Camera	Zhiwei D132s
Six-Axis Force Sensor	Desenter γ82
Ultrasound Probe	Wisonic Clover60
Main Computing Hub	NVIDIA Jetson Nano-B01

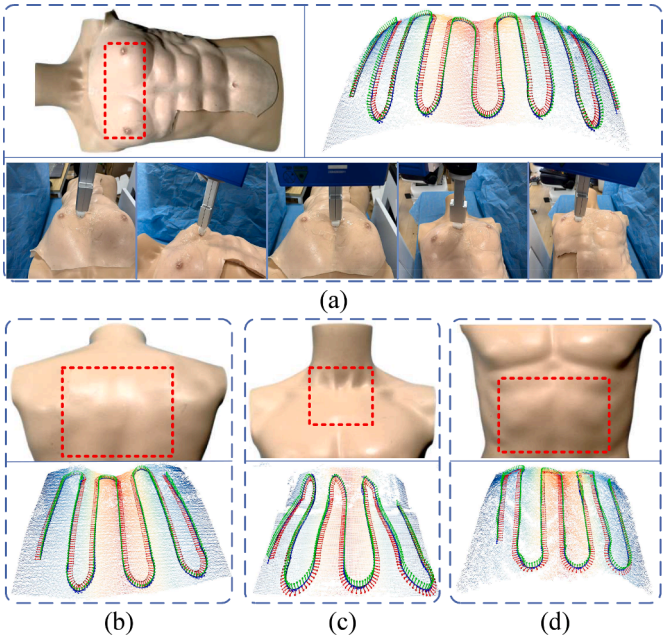


Fig. 18. Trajectory generation results in different regions. (a) Chest; (b) Back; (c) Neck; (d) Abdomen.

of the key hardware components used in the system. The overall control architecture is implemented in the Robot Operating System 1 (ROS 1) platform.

5.2. Scanning trajectory planning

By setting $W_{\text{probe}} = 25$ mm and $W_{\text{overlap}} = 5$ mm, the scanning trajectories generated for various anatomical regions of the human body are depicted in Fig. 18.

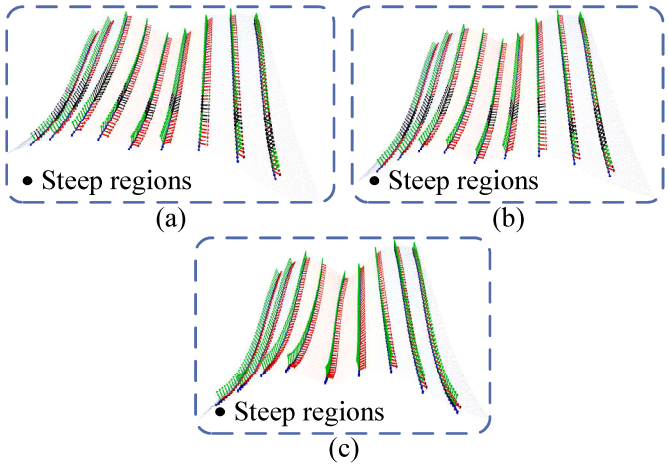


Fig. 19. Trajectory generation results of different algorithms. (a) 2D projection method; (b) Planar slicing method; (c) Proposed method.

Table 4
Performance comparison of different trajectory planning algorithms.

Method	COV (%)	E_s (mm)	$\bar{\theta}$ (°)	$\theta_{\max}$ (°)	T_{ce} (s)
2D Projection	90.5226	1.0257	0.7102	2.5377	1.3921
Planar Slicing	97.3521	0.8105	0.7533	3.2321	2.0769
Proposed	99.1663	0.6119	0.5092	1.3585	1.5933

To demonstrate the superiority of the proposed trajectory planning method, two approaches are selected as baselines for comparison: the 2D projection method (Welleweerd et al., 2020) and the planar slicing method (Tan et al., 2022). The following four metrics are employed for evaluation:

Coverage completeness: A point is covered when its minimum Euclidean distance to the trajectory is less than or equal to Δd. The coverage rate is:

$$\text{COV}(\%) = \frac{N_c}{N_t} \times 100\% \tag{43}$$

where N_c and N_t denote the numbers of covered and total points, respectively.

Spacing uniformity: The trajectory spacing is evaluated by the RMSE:

$$E_s = \sqrt{\frac{1}{K}\sum_{i=1}^{K}(d_i - \bar{d})^2} \tag{44}$$

where d_i is the minimum distance to the adjacent trajectory and $\bar{d}$ is its mean value.

Pose smoothness: The normal angular deviation (NAD) between adjacent points is calculated as:

$$\theta_i = \arccos(\mathbf{n}_i \cdot \mathbf{n}_{i+1}) \tag{45}$$

where $\mathbf{n}_i$ and $\mathbf{n}_{i+1}$ are adjacent unit normal vectors. The mean and maximum NAD are used for evaluation, and $\theta_i \geq 2°$ indicates steep regions.

Computational efficiency: The trajectory planning time T_{ce} is recorded.

The comparative experiments of trajectory generation methods are conducted on the lumbar phantom. To facilitate performance comparison, the trajectory connection strategy proposed in this paper is omitted, and the same preprocessing and trajectory smoothing strategies are adopted across all methods. Fig. 19 shows the trajectories generated by the three algorithms, and the detailed performance metrics are summarized in Table 4.

First, the proposed method achieves a coverage rate of 99.17% and a spacing RMSE of 0.61 mm, outperforming both baseline methods. This

Table 5
Control parameter configurations for the four controllers in experiments.

Controller	Parameter Settings
AC	$m = 3.5$, $b = 1000$, $k = 0$
AAC	$\alpha = 0.01$, $\beta = 1.2$
DOBSMAC	$C = 3$, $k_s = 11$, $\eta = 5$, $L_1 = 50$, $L_2 = 150$
DO-MPAC	$k_{h0} = 1500$, $N_p = 15$, $Q = 0.5$, $R = 10$, $\omega_o = 8$

performance stems from constructing a geodesic distance field directly on the 3D surface and extracting isolines, which preserves the topological features of complex anatomies. In contrast, the 2D projection method (COV: 90.52%) suffers from coverage blind spots and 3D physical spacing distortions during 2D-to-3D mapping, particularly in high-curvature regions. Meanwhile, the planar slicing method (COV: 97.35%) faces an inherent trade-off: larger slicing intervals lose local details, whereas smaller intervals increase the computational burden of intersection calculations. Second, the proposed method yields the lowest mean NAD ($\bar{\theta} = 0.51°$) and maximum NAD ($\theta_{\max} = 1.36°$). Notably, both baseline methods produce steep regions ($\theta_{\max} > 2°$), which can trigger abrupt probe pose shifts and contact force instability. By introducing anisotropic weights during the graph search to heavily penalize abrupt normal variations, the proposed method suppresses drastic pose transitions. Regarding computational efficiency (T_{ce}), although the 2D projection method is the fastest (1.39 s) due to the omission of 3D topological calculations, it sacrifices overall trajectory quality. The proposed method (1.59 s) is computationally more efficient than the planar slicing method (2.08 s). This efficiency is attributed to the multi-scale spatial discretization strategy, which reduces the computational burden while preserving high-precision 3D topological features.

5.3. Comparative force tracking performance

The physical experiments are conducted in two distinct scenarios. **Experiment 1** utilizes the chest phantom illustrated in Fig. 18(a), which features a spatially complex curved surface with varying stiffness. Specifically, the synthetic muscle layer is thinner in the center and thicker at the lateral margins, yielding a heterogeneous stiffness profile across the phantom. In this setup, the proposed trajectory planning algorithm generates a predefined path that satisfies the probe pose constraints, followed by the execution of the force control experiments. Furthermore, to verify the robustness of the proposed DO-MPAC, **Experiment 2** is conducted on an unknown curved surface featuring abrupt stiffness transitions, as depicted in Fig. 20. In this scenario, the probe pose constraints are relaxed, and the probe translates along a single direction while maintaining a vertical orientation. The surface comprises two materials configured to exhibit lower stiffness on the two sides and higher stiffness in the middle. During the scanning procedure, the probe relies entirely on the admittance control algorithm to adapt to geometric variations while traversing the low-high-low stiffness sequence. This uncertain environment simulates unmodeled physiological disturbances to evaluate the robustness of the different controllers.

Identical baseline admittance parameters, namely $m = 3.5$, $b = 1000$, and $k = 0$, are applied to all four evaluated controllers. The detailed parameter configurations for the respective control strategies are provided in Table 5. For a fair comparison, the parameters of all controllers are derived from the simulations and their original literature, then fine-tuned to perform optimally in **Experiment 1**.

During the experiments, the ultrasound probe moves at a constant $v = 3.5$ mm/s while the admittance controllers maintain the target force despite environmental variations. To ensure safe interaction, the physical saturation constraints are set to $|u_0| \leq 500$ and $|\Delta x| \leq 20$ mm, based on the analysis in Remark 4. At the hardware level, the control and data sampling periods are 5 ms and 1 ms, respectively. Finally, to reduce random errors, experiments for each parameter set are repeated five times under identical conditions, with the results averaged.

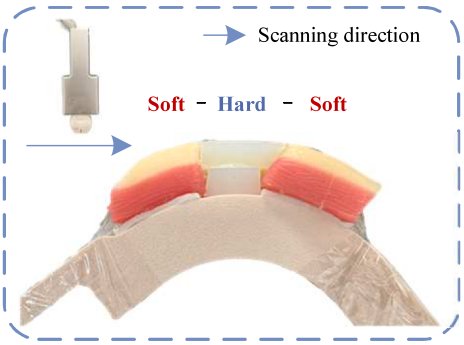


Fig. 20. Robustness evaluation on a complex curved surface.

Table 6
Quantitative experimental performance metrics of the evaluated controllers.

Experiment	Method	MFD (%)	IAE (N·s)	ITAE (N·s^2)	HFE (N^2)
Experiment 1	AC	91.6932	29.8304	259.6152	0.0137
	AAC	73.0132	32.9089	250.2553	0.0131
	DOBSMAC	107.1534	29.0152	223.5631	0.0138
	DO-MPAC	11.4342	19.9953	151.6846	0.0092
Experiment 2	AC	246.8730	102.7705	1860.3513	0.0667
	AAC	254.4134	73.7892	1126.2031	0.0361
	DOBSMAC	209.6732	55.2853	740.3774	0.0327
	DO-MPAC	24.9383	30.0884	421.7086	0.0149

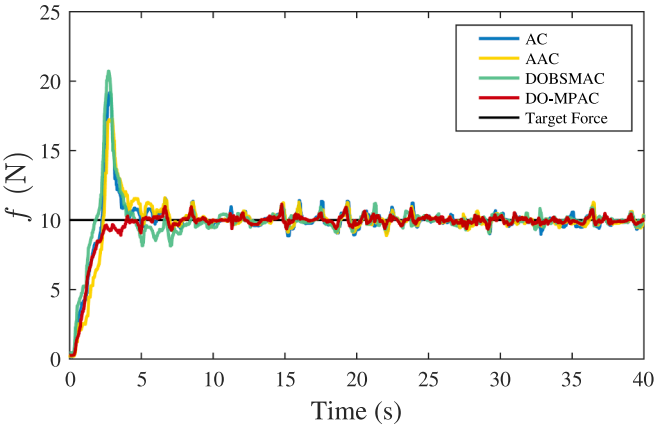


Fig. 21. Comparison of contact force tracking results in **Experiment 1**.

The force tracking results obtained from the chest phantom are illustrated in Figs. 21–23, and the corresponding quantitative performance metrics of the four control methods are compared in Table 6.

Fig. 21 presents the force responses of the different controllers. The AC, AAC, and DOBSMAC strategies initially reach the target force at approximately 2 s. However, they exhibit MFD of 91.69%, 73.01%, and 107.15%, respectively, during the initial convergence phase. Although the proposed DO-MPAC requires approximately 4 s to reach the target force, it nears the desired 10 N level around 2 s while maintaining an MFD of 11.43%. Regarding the overall tracking error, the IAE and ITAE of DO-MPAC are 20.00 N · s and 151.68 N · s^2, respectively, which are the lowest among the four controllers. The box plots in Fig. 22 illustrate force fluctuations across the five independent tests. This visual comparison confirms high repeatability, justifying the use of averaged results to mitigate random noise. Furthermore, Fig. 23 plots the evolution of the lumped disturbance $\hat{D}(t)$ estimated by the ESO and the cost function J. As observed, the trends of these two variables align closely with the

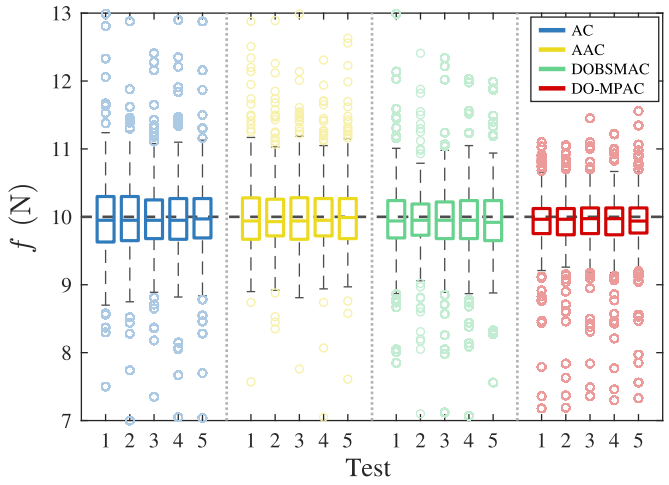


Fig. 22. Force tracking fluctuations of different controllers in **Experiment 1**.

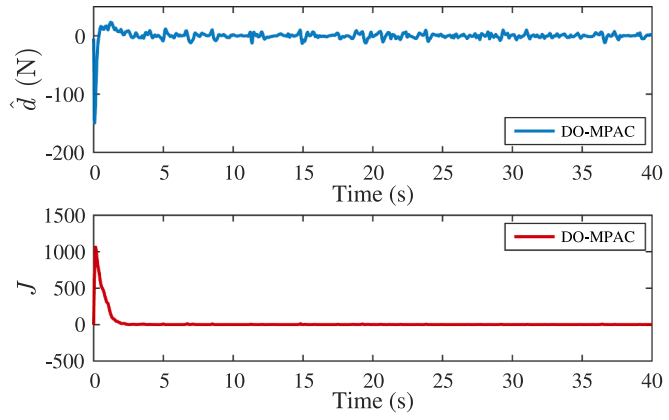


Fig. 23. Estimated disturbance $\hat{D}(t)$ and the cost function J in **Experiment 1**.

force data. On the chest phantom, the dynamic uncertainties are accurately tracked, which facilitates the steady-state stability of DO-MPAC.

Overall, while all four controllers successfully achieve the target force on the chest phantom, their transient behaviors differ significantly. Although AAC and DOBSMAC exhibit improved steady-state accuracy and faster response compared to the baseline AC, they still suffer from prominent initial overshoots. Such excessive contact forces pose potential safety risks during human-robot interaction tasks like ultrasound scanning. Conversely, the proposed DO-MPAC effectively suppresses these initial peaks, delivering a smooth and safe convergence. As reflected by its minimal MFD, IAE, and ITAE values, DO-MPAC provides the most balanced and reliable tracking performance in this scenario.

The experimental results for the ultrasound scanning on the unknown curved surface are presented in Figs. 24–26, with the corresponding performance metrics compared in Table 6.

Fig. 24 presents the force responses in **Experiment 2**. The AAC strategy exhibits the slowest transient response, reaching the target force at approximately 2.1 s, whereas AC and DOBSMAC initially reach this target around 1.2 s. During the initial convergence phase, DO-MPAC yields an overshoot of approximately 9.89%. In contrast, the AC, AAC, and DOBSMAC strategies exhibit significant initial overshoots of 246.87%, 254.41%, and 209.67%, respectively. Notably, between 15 s and 30 s, the probe traverses the stiffer region of the surface, which is characterized by higher stiffness. In this area, the AC method experiences pronounced force oscillations, resulting in tracking degradation. Conversely, DO-MPAC maintains stable force variations throughout this phase. As indicated in Table 6, the IAE and ITAE of DO-MPAC are 30.09 N · s and 421.71 N · s^2, respectively, which are the lowest among the four controllers. The force fluctuations across the five independent tests, plotted in Fig. 25, further demonstrate the improved tracking precision and robustness of DO-MPAC. Additionally, the evolution of the estimated disturbance $\hat{D}(t)$ and the cost function J depicted in Fig. 26 confirms the adaptive capability of the designed scheme. Specifically, owing to the surface undulations and spatially varying stiffness of the surface, the cost function J exhibits corresponding dynamic adjustments during the steady-state phase. Notably, DO-MPAC achieves the lowest HFE in both **Experiment 1** and **Experiment 2**, which is attributed to the MPC cost function design that effectively smooths the control inputs.

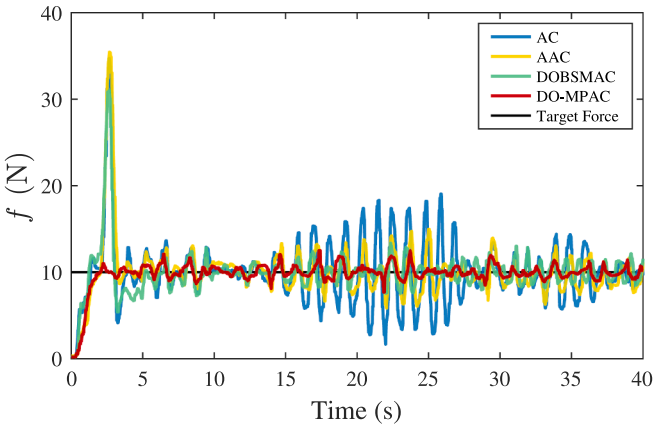


Fig. 24. Comparison of contact force tracking results in **Experiment 2**.

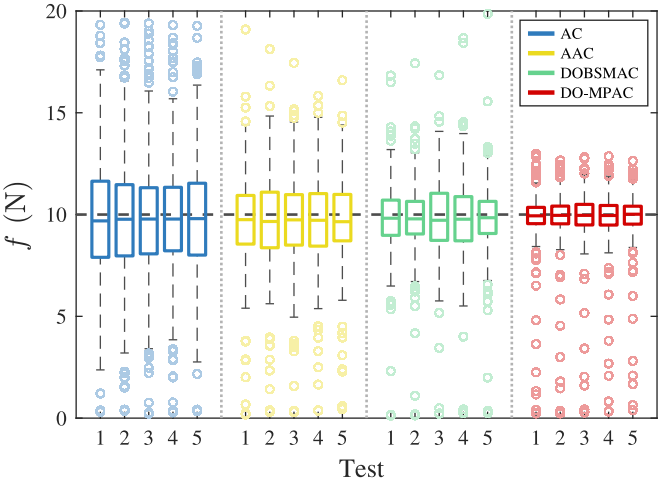


Fig. 25. Force tracking fluctuations of different controllers in **Experiment 2**.

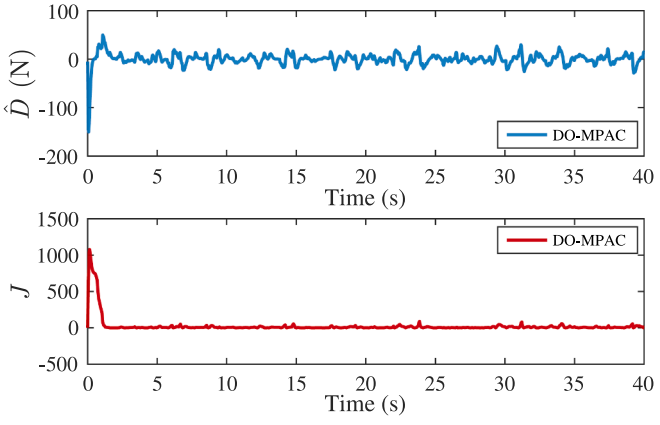


Fig. 26. Estimated disturbance $\hat{D}(t)$ and the cost function J in **Experiment 2**.

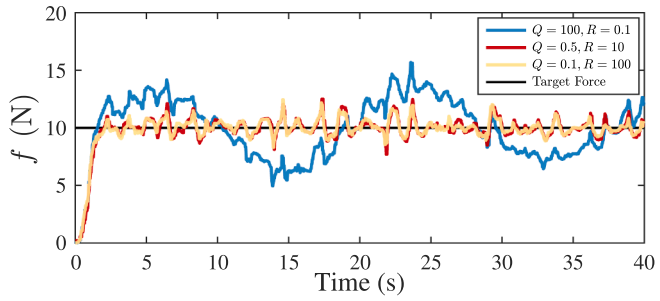


Fig. 27. Parameter sensitivity to MPC weights (Q, R).

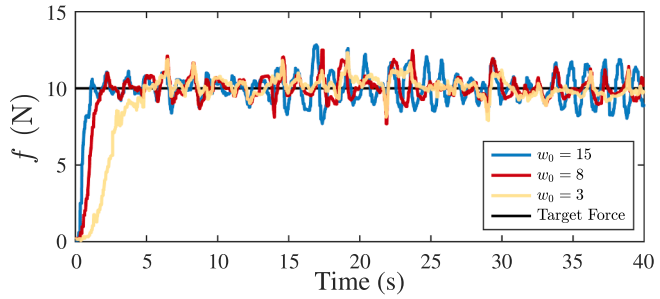


Fig. 28. Parameter sensitivity to ESO bandwidth (ω_o).

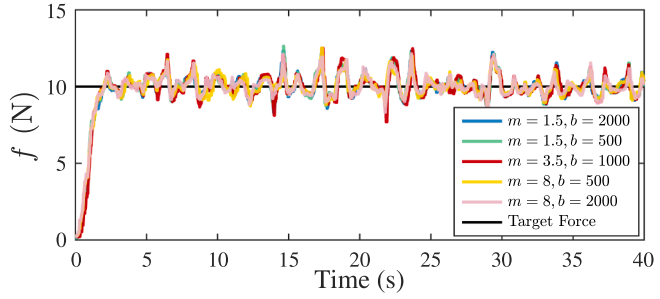


Fig. 29. Force response of DO-MPAC with different admittance parameters (m, b).

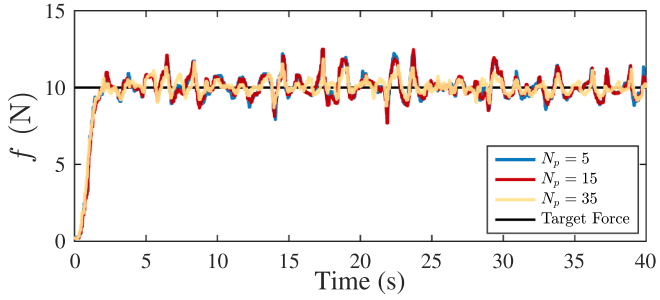


Fig. 30. Force response of DO-MPAC with different N_p.

In summary, the comprehensive evaluations across both scenarios reveal the inherent limitations of the baseline methods. The conventional AC with fixed parameters is susceptible to significant force fluctuations when environmental conditions change. Although the parameter adaptation mechanism of AAC improves its adaptability, its reliance on error accumulation results in substantial transient force errors during stiffness transitions. By integrating a disturbance observer with SMC, DOBSMAC estimates and partially mitigates environment-induced disturbances. However, its performance remains constrained by the fixed reaching dynamics of the linear sliding surface. Furthermore, as observed in the experiments, the discontinuous switching nature of SMC induces inherent chattering, which manifests as high-frequency force vibrations during contact. To overcome these limitations, the proposed DO-MPAC leverages receding horizon optimization to plan optimal force trajectories predictively rather than relying solely on reactive error compensation. By explicitly handling dynamic constraints and anticipating environmental variations, it consistently maintains high-quality interaction. Ultimately, it outperforms the other evaluated controllers across all tracking metrics, thereby validating its effectiveness and adaptability in complex interaction environments.

5.4. Parameter sensitivity and real-time feasibility

Furthermore, parameter sensitivity tests in **Experiment 2** evaluate the impact of Q, R, and ω_o. Figs. 27-28 display the results, and Table 7 summarizes the quantitative metrics.

It can be observed that increasing the Q/R ratio to accelerate the tracking response (e.g., $Q = 100$, $R = 0.1$) leads to degraded steady-state performance. Specifically, the system exhibits low-frequency fluctuations and deviates from the target force, which produces the highest MFD, IAE, and ITAE values. This occurs because a high Q makes the controller overly sensitive to force errors, and a low R inadequately penalizes the control input. Consequently, the system overreacts to sensor noise and tissue micro-deformations. These aggressive control commands excite unmodeled robot dynamics, which causes continuous overcompensation instead of convergence. Conversely, the configuration with $Q = 0.1$ and $R = 100$ delivers superior steady-state performance and the lowest ITAE value. A large R penalizes input variations, which generates smoother and more conservative motion commands. Thus, the probe maintains stable tissue compliance and minimizes steady-state errors.

The experimental data for the ESO bandwidth ω_o highlights a trade-off between theoretical expectations and practical constraints. Theoretically, a higher ω_o speeds up disturbance estimation, which accelerates the transient response. However, in physical experiments (e.g., $\omega_o = 15$), an excessively high bandwidth amplifies high-frequency noise and structural vibrations. This noise enters the feedforward compensation term z_3 as false disturbances, which causes minor steady-state chattering and degrades the error metrics compared to the $\omega_o = 8$ case. Conversely, a lower bandwidth (e.g., $\omega_o = 3$) provides stronger low-pass filtering. Although this slows the transient response, it filters out high-frequency interference and yields a smooth disturbance estimate. Therefore, the DO-MPAC tuning process requires a balance between rapid response and steady-state stability. This trade-off explains the discrepancy between the optimal bandwidths in the simulation ($\omega_o = 90$) and the physical experiments ($\omega_o = 8$).

Notably, the force response curves show no significant overshoot across all Q, R, and ω_o configurations, which demonstrates the safety and robustness of DO-MPAC. The ESO feedforward compensation effectively suppresses the influence of nonlinear contact dynamics, and the MPC receding horizon optimization constrains abrupt state variations. Together, these mechanisms improve safe human-robot interaction.

To verify the DO-MPAC parameter low-dependency analysis in Remark 3, five sets of m and b are tested in **Experiment 2**. Fig. 29 shows the results, and Table 7 lists the metrics. Despite wide parameter variations, the force response curves show similar initial convergence and consistent steady-state tracking. Even when the virtual mass m (1.5 to 8) and damping b (500 to 2000) change significantly, all error metrics remain within a narrow range. The minor numerical differences reflect actual physical interaction dynamics. Real contact environments inevitably introduce unmodeled impacts, which cause a brief ESO phase delay. During this delay, m and b act as physical impedance, which absorbs the residual high-frequency energy. Overall, these results demonstrate that DO-MPAC effectively suppresses the traditional tuning trade-off between compliance and rapid response, which proves its strong parameter robustness.

Table 7
Quantitative performance metrics of DO-MPAC under different parameter settings.

Parameters	Values	MFD (%)	IAE (N·s)	ITAE (N·s^2)
Weights (Q, R)	$Q = 100, R = 0.1$	56.6887	87.5671	1566.3672
	$Q = 0.5, R = 10$	24.9383	30.0884	421.7086
	$Q = 0.1, R = 100$	24.4035	27.0056	352.8618
Bandwidth (ω_o)	$\omega_o = 15$	28.3538	33.3919	633.8973
	$\omega_o = 8$	24.9383	30.0884	421.7086
	$\omega_o = 3$	23.1583	40.2548	383.4374
Admittance (m, b)	$m = 1.5, b = 2000$	23.1732	28.6681	404.2125
	$m = 1.5, b = 500$	26.2889	27.1199	368.1447
	$m = 3.5, b = 1000$	24.9383	30.0884	421.7086
	$m = 8, b = 500$	25.1312	26.4574	358.9115
	$m = 8, b = 2000$	21.3303	26.2377	367.8479
Horizon (N_p)	$N_p = 5$	24.0488	29.9249	424.9066
	$N_p = 15$	24.9383	30.0884	421.7086
	$N_p = 35$	18.5403	22.7586	275.1407

Table 8
Quantitative computational time metrics of the evaluated controllers.

Method	Parameter	Mean (μs)	Max (μs)	Ratio > 5000μs (%)
AC		0.6282	130.2640	0.0000
AAC		0.6460	289.9560	0.0000
DOBSMAC		1.2795	305.1640	0.0000
DO-MPAC	$N_p = 5$	42.2618	15139.2000	0.0325
	$N_p = 15$	43.1690	22888.5000	0.0300
	$N_p = 35$	43.7656	22565.9000	0.0425

Finally, computational time tests are conducted to evaluate the real-time feasibility. Table 8 compares the timing metrics across the four controllers. Unlike AC, AAC, and DOBSMAC with explicit control laws, DO-MPAC incorporates a predictive optimization framework, which increases its computational time. However, due to the offline precomputation of the analytical MPC gain matrix discussed in Section 3.3, its average time stays around 43μs. Rare cycles exceed the 5000μs limit, but the overtime ratio remains below 0.1%. For these overtime cycles, the system applies a zero-order hold to maintain the previous control command, which ensures continuous and safe operation. These data show that DO-MPAC meets real-time requirements over 99.9% of the time, and its computational burden fits within hardware limits. Furthermore, various N_p values are tested in **Experiment 2** to evaluate prediction horizon impacts. Fig. 30 shows the results, and Table 7 lists the metrics. It can be observed that N_p creates a clear trade-off between computational cost and control accuracy. A larger N_p gives the MPC a broader future vision. The controller evaluates long-term impacts over a wider time window, which prevents short-sighted suboptimal decisions. Consequently, the probe plans smoother and more forward-looking motion commands, which achieves better global tracking accuracy.

In summary, the integration of the disturbance observer and MPC effectively meets the constant force control requirements for ultrasound probes in complex soft tissues. The ESO estimates and compensates for lumped disturbances, which reconstructs the force error into a nominal second-order system. Consequently, the virtual parameters m and b operate independently of nominal tracking. They act as physical impedance, which absorbs high-frequency impacts during initial contact or ESO delays. The MPC weights Q and R govern nominal responses. A higher Q/R ratio accelerates transient tracking but amplifies noise, which causes steady-state fluctuations. Conversely, a larger R generates smoother commands, which improves steady-state accuracy. Furthermore, the ESO bandwidth ω_o determines feedforward compensation. A larger ω_o speeds up estimation but introduces measurement noise. A smaller ω_o provides low-pass filtering, which benefits steady-state performance but delays the transient response.

Based on these mechanisms, practical DO-MPAC tuning follows four steps. First, define the theoretical boundaries $[\omega_{\min}, \omega_{\max}]$ using disturbance frequencies and hardware limits. For physical deployment, select an initial ω_o based on simulation results to ensure baseline observation. Second, adjust Q and R to ensure stable steady-state compliance without low-frequency deviations, even if this delays transient convergence. Third, increase ω_o to compensate for the transient delay, which improves rapid response and maintains steady-state stability. Finally, iterate between the MPC weights and ω_o until the system achieves an optimal balance between transient speed and steady-state accuracy.

6. Conclusion and discussion

This paper has presented an integrated autonomous compliant ultrasound scanning framework based on 3D point clouds to maintain consistent probe poses and stable contact forces on complex human surfaces. For trajectory planning, an anisotropic geodesic distance field is established to adapt to various anatomical topologies, which satisfies probe pose constraints and avoids abrupt orientation shifts. Regarding force control, the proposed DO-MPAC scheme is developed to maintain a stable and compliant interaction force. By combining an extended state observer (ESO) with receding horizon optimization, this strategy effectively suppresses environmental uncertainties and predictively ensures optimal contact force transitions. Additionally, the formulation of a virtual control law structurally suppresses the coupling effect between the force tracking dynamics and the admittance parameters. This scheme mitigates the traditional trade-off between physical compliance and tracking agility. Simulations and physical experiments have confirmed that DO-MPAC improves upon conventional reactive methods in transient response, steady-state precision, and robustness against varying stiffness. Specifically, in highly curved environments with variable tissue stiffness, DO-MPAC achieves rapid force tracking while mitigating the transient force overshoots that compromise interaction safety in traditional control architectures. Ultimately, this research addresses the performance limitations of existing methods in time-varying contact environments, offering a robust and clinically viable solution for robot-assisted ultrasound examinations.

Despite the demonstrated capabilities of the proposed framework, several limitations remain to be addressed. From a perception and planning perspective, the current trajectory generation relies on static, high-quality 3D point clouds, making the system susceptible to real-time sensor noise, occlusions, and dynamic tissue deformations. In terms of control and execution, while the simplified 1D normal admittance model effectively preserves scanning reproducibility, the effects of tangential motion and multi-axis coupling warrant more explicit consideration. Additionally, while the current analytical MPC ensures computational efficiency, it omits explicit online constraints. Incorporating explicit constrained optimization without compromising the high-frequency control loops remains a computational challenge for future development. Finally, although DO-MPAC exhibits strong robustness in complex curved environments, its performance under sudden patient movements or atypical physiological disturbances requires rigorous validation. Future work will focus on incorporating real-time dynamic trajectory updates and explicit multi-axis dynamic modeling. To facilitate clinical translation, future studies will include *in vivo* validation on human subjects, where the system's robustness against complex physiological variations and unmodeled disturbances can be further evaluated, thereby improving its clinical feasibility.

CRediT authorship contribution statement

Jiaen Shen: Writing – original draft, Validation, Software, Methodology, Conceptualization; **Yongfei Feng:** Writing – review & editing, Supervision, Formal analysis; **Qiqi Pan:** Writing – review & editing, Software, Formal analysis; **Yuhui Cui:** Writing – review & editing, Software, Formal analysis; **Jingcong Zhang:** Writing – review & editing, Formal

analysis; **Hongbo Wang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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